LAB | AWS Security Extra

Skills you will learn

Cloud Computing: Cloud Fundamentals

Introduction

Security is a critical aspect of managing AWS resources. In this lab, you will explore essential AWS security services and understand how to implement a **layered security approach** (also known as Defense in Depth). You will configure Identity and Access Management (IAM), monitor security threats, and encrypt sensitive data.

By the end of this lab, you will:

- Understand key AWS security services and their use cases.
- Implement IAM roles, policies, and multi-factor authentication (MFA).
- Configure AWS security monitoring tools like GuardDuty and CloudTrail.
- Apply encryption using AWS Key Management Service (KMS).

Prerequisites

- **AWS Account**: Ensure you have administrative access.
- **Basic Knowledge**: Familiarity with AWS IAM, encryption, and networking concepts.
- **AWS CLI (Optional)**: If you prefer command-line interactions.

Step 1: Identity and Access Management (IAM)

1.1 Understanding IAM

- **IAM Users and Groups**: Assign least-privilege access to users.
- **IAM Roles**: Grant permissions to AWS services (e.g., EC2 accessing S3).
- **IAM Policies**: Define permissions using JSON-based rules.

1.2 Configure IAM User and MFA

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1. **Go to IAM in AWS Console**.
2. Click on **Users** > **Add User**.
   - **User Name**: `DevOpsAdmin`
   - **Access Type**: AWS Management Console access.
3. **Set Permissions**:
   - Attach **AdministratorAccess** policy (for testing purposes).
4. **Enable MFA**:
   - Go to **Security Credentials** and enable **Virtual MFA**.
   - Scan the QR code with an authenticator app (Google Authenticator, Authy).
   - Enter the two generated codes to confirm.
5. **Test IAM Login**:
   - Log out and sign in as `DevOpsAdmin` to confirm access.

Step 2: Encryption with AWS KMS

2.1 Create an Encryption Key

1. Go to **AWS Key Management Service (KMS)**.
2. Click **Create Key** > Select **Symmetric Key**.
3. Configure key settings:
   - **Alias**: `MySecureKey`
   - **Key Administrators**: Select your IAM user.
   - **Key Users**: Allow EC2 instances to use this key.
4. **Create Key** and save the Key ID.

2.2 Encrypt S3 Data with KMS

1. Go to **S3** > Create a new bucket.
2. Under **Default Encryption**, select **AWS KMS Key (SSE-KMS)**.
3. Choose **MySecureKey**.
4. Upload a test file and confirm encryption settings.

Step 3: Monitor AWS Security Events

3.1 Enable AWS CloudTrail

1. Navigate to **AWS CloudTrail**.
2. Click **Create Trail** > Select **Management & Data Events**.
3. Enable **S3 and Lambda event logging**.

4. Choose an S3 bucket for log storage.
5. Click **Create** and monitor logs under CloudTrail.

3.2 Activate Amazon GuardDuty

1. Go to **AWS GuardDuty**.
2. Click **Enable GuardDuty**.
3. Monitor for findings related to suspicious activity.
4. Test an alert by accessing an AWS service from an unusual region.

Step 4: Implement Web Application Security

4.1 Deploy AWS WAF

1. Navigate to **AWS WAF & Shield**.
2. Click **Create Web ACL** > Select **CloudFront or ALB**.
3. Add a **rule** to block known malicious IP addresses.
4. Attach the ACL to an application load balancer (if applicable).
5. Test the WAF by attempting an SQL injection or XSS attack.

4.2 Use AWS Secrets Manager

1. Go to **AWS Secrets Manager**.
2. Click **Store a new secret**.
3. Choose **Credentials for RDS Database**.
4. Enter a mock username and password.
5. Store and retrieve secrets programmatically using the AWS SDK.

Summary

By completing this lab, you have:

- Configured **IAM users, roles, and MFA** to control access securely.
- Used **AWS KMS** to encrypt data stored in S3.
- Enabled **AWS CloudTrail and GuardDuty** for security monitoring.
- Implemented **AWS WAF** for web application security.
- Stored sensitive credentials using **AWS Secrets Manager**.

AWS security follows a **layered approach**, ensuring protection at multiple levels, from identity management to encryption, monitoring, and application security. 🚀

Submission Guidelines

- Take **screenshots** of each major step:
  - IAM user creation (`DevOpsAdmin`) and **MFA setup confirmation**.
  - **KMS key creation** (`MySecureKey`) and **S3 bucket encryption** settings.
  - Uploaded file in S3 showing **SSE-KMS encryption** with your key.
  - **CloudTrail trail setup** and evidence of event logging.
  - **GuardDuty enabled** with example findings or confirmation screen.
  - **Web ACL configuration** in AWS WAF with at least one rule (e.g., IP block).
  - New secret stored in **AWS Secrets Manager** (show name and type only).
- Save all screenshots into a **document (Google Docs, PDF, etc.)**.
- Upload the document to **Google Drive** and **share the link**.
- Ensure your **instructor has viewing access** before submission.

Next Steps

- **Test IAM Policies**: Create a least-privilege IAM role.
- **Enable AWS Config**: Track configuration changes for compliance.
- **Try Incident Response**: Simulate a security event using AWS Lambda automation.

Keep experimenting to strengthen your AWS security skills!

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